Lakshman Marapelli

Data Scientist

EDUCATION

Data Science And Gen-AI, *Innomatics Research Lab* 2026
Course work in Python, EDA, MySQL, Power BI, Statistics, ML, DL, Gen AI, NLP, Computer Vision

Bachelor of Science (MSCS), *Government City College(A)* 2025
Mathematics, Statistics and Computer Science

SKILLS

Python
Python: Anaconda, Jupyter Notebook, Google Colab, Syntax, Identifiers & Operators, Arrays, NumPy, Data Manipulation, Pandas

Power BI
Power BI workflow, Visualization, Trend Data viz, Power Queries, Power Pivot, DAX, DA Expression, Web & RLS, Visual Interactions, Drill Through

Advanced Statistics
Descriptive Statistics, Probability Distributions, Data Collection Techniques, Inferential Statistics

SQL + Power BI
MySQL, NoSQL, CRUD Operations, RDBMS, Data Exploration and Filtering, DQL Operators, Clauses, Joins, ACID, COMMIT, ROLLBACK

EDA (Exploratory Data Analysis)
Data Profiling, Data Cleaning, Missing Value & Outlier Handling, Descriptive Statistics, Univariate-Bivariate- Multivariate Analysis, Correlation Analysis, Feature Understanding, Data Transformation, Pattern & Trend Identification, Insight Generation

PROJECTS

Grocery Management System Using SQL
- Designed and implemented a **normalized relational database** for a grocery store using **multiple tables, primary keys, and foreign key constraints**, managing **1000+ simulated transaction records**.
- Wrote **15+ advanced SQL queries** using **JOINs, GROUP BY, subqueries, and aggregate functions** to analyze sales performance and operational metrics.
- Generated insights on the **top 10 customers, best-selling products, supplier contribution, and monthly order trends** from transactional data.
- Improved data retrieval efficiency by **structuring queries and relationships across 6+ relational tables**.
- Presented **data-driven insights and business recommendations** to stakeholders using summarized query outputs

Hyderabad Real Estate Market Analysis (Web Scraping + EDA using Python)
- Scraped **2000+ residential property listings** from a real estate website using **Selenium, BeautifulSoup, and Regex** for structured data extraction.
- Performed **data cleaning and preprocessing on 10+ variables** using **Python (pandas, NumPy),** including missing value handling, formatting, and feature engineering.
- Conducted **Exploratory Data Analysis (EDA)** and **ANOVA hypothesis testing** to evaluate property pricing trends across **multiple locations and property types**.
- Built **10+ visualizations** using Python to identify patterns in price distribution, locality demand, and property characteristics.
- Delivered insights that helped identify **high-demand residential areas and key price-driving factors**.

EV-Sales Data Set (Power BI Project)
- Imported and transformed **10,000+ EV sales records** using **Power Query** to prepare structured data for analysis.
- Created **10+ DAX measures and calculated columns** to track KPIs such as **total sales, growth rate, and category performance**.
- Built **interactive dashboards with 6+ visualizations** to monitor EV adoption trends across regions and time periods.
- Analyzed sales patterns to identify **top-performing vehicle categories and seasonal demand trends**.
- Presented **data-driven insights through dynamic dashboards,** enabling easier stakeholder decision-making.

Virtual Mouse using OpenCV
- Developed a **gesture-based virtual mouse system** using **OpenCV** and **MediaPipe**, enabling **5 mouse actions** (cursor movement, left click, right click, double click, scrolling).
- Implemented **21-point hand landmark tracking** for real-time gesture recognition with **~25–30 FPS performance**.
- Designed gesture detection logic using **finger angle calculation, distance metrics, and landmark mapping** for reliable control.
- Integrated **PyAutoGUI** and **pynput** to convert gestures into system-level mouse commands.
- Improved usability by implementing **cursor smoothing, gesture debouncing, and scroll-zone mapping** to reduce unintended clicks and stabilize cursor movement.

CERTIFICATES

- Data Analysis with MYSQL by Innomatics Research Labs
- Python programming by Innomatics Research Labs
- PowerBI completion by Innomatics Research Labs
- EDA using Python by Innomatics Research Labs

LANGUAGES

- English
- Hindi
- Telugu